PORTALDOT HACKER HOUSE 2026 · TRACK 3: NODE OPERATIONS & TOOLING

PortaPulse

High-performance node operations and real-time telemetry for the Portaldot network.

Live RPC data

Read straight from wss://mainnet.portaldot.io

Four operator tools

Telemetry, RPC benchmark, batch planner, heartbeat alerts

Open source

MIT licensed, React 19 + @polkadot/api, no mocks

naomi197.github.io/portapulse

THE PROBLEM

Node operators fly blind on Portaldot

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01 No native telemetry view

Validators and node runners have no single place to watch block height, finality lag, peer count, or head size for a Portaldot node.

02 Slow blocks hide failures

Portaldot produces roughly one block per minute. A stalled chain looks identical to a quiet one until it is far too late to react.

03 RPC endpoints are a black box

There is no quick way to compare latency or health between a public endpoint and a local development node before trusting it.

04 Outages are noticed after the fact

Without an alerting path, desync, peer isolation, and finality stalls surface only when someone happens to look.

THE SOLUTION

One dashboard that reads the chain directly

Real-time telemetry

Latest block, median block time, extrinsics in head, active peers, and finalized head, refreshed from live RPC.

RPC benchmark

Timed health checks across public and local endpoints so operators can pick the fastest path.

Batch load planner

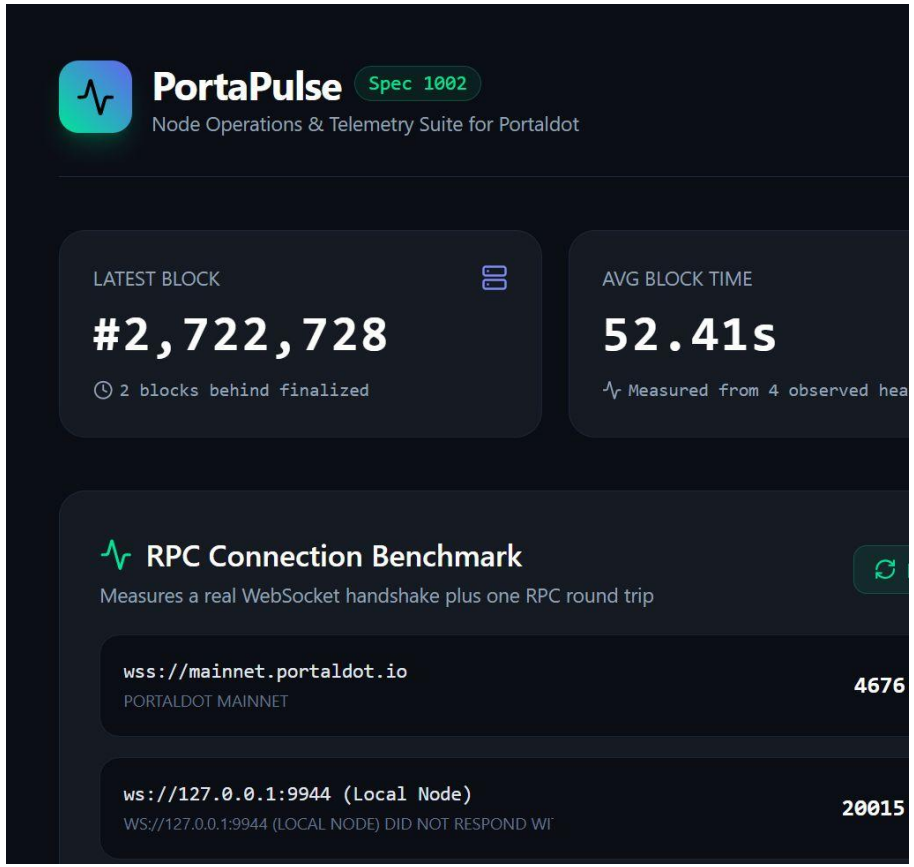
Sizes a transaction batch against the observed head size and reports estimated blocks to clear.

Heartbeat alerts

A standalone worker polls the chain and posts real alerts to Discord or Slack when a node degrades.

EVIDENCE — LIVE DATA

Every metric is a real RPC value, not a mock



Observed against wss://mainnet.portaldot.io

Chain	**Portaldot Mainnet**
Client	**Portaldot Node 2.0.0**
Runtime spec	**1002**
Token	**POT · 14 decimals**
SS58 prefix	**42**
Block production	**~1 block / minute**

Independent checks ship with the repo: **verify-client.mjs** and **measure-block-time.mjs** connect to the live network and print the same numbers.

OPERATIONS — ALERTING

The worker catches failures the browser cannot

A browser tab cannot wake you when your machine is asleep. **worker/heartbeat.mjs** is a standalone Node process that polls the chain and delivers real alerts.

Condition	Severity	Default threshold
No new block for a while	Critical	180 s
Node reports it is still syncing	Warning	—
Peer count below the minimum	Critical	1 peer
Finality lag grows too large	Warning	10 blocks

Alerts fire once when a condition opens, repeat on a cooldown while it persists, and send a recovery notice when it clears.

WHY PORTALDOT

Node health is the product's whole premise

Portaldot produces blocks slowly

Roughly one block per minute means a stall is easy to miss and expensive to diagnose after the fact. PortaPulse treats finality lag and block silence as first-class signals.

Operator tooling is the gap

Portaldot V3 combines Substrate and EVM execution. Validators and infrastructure teams need monitoring, benchmarking, and alerting that speaks that stack directly.

Built on Portaldot primitives

PortaPulse reads Portaldot RPC through @polkadot/api, targets the Portaldot Mainnet and local development endpoints, and reports Portaldot chain properties — POT, SS58 prefix 42, runtime spec 1002.

STATUS & NEXT STEPS

Working today, with a clear runway

Shipped

- Live dashboard reading Portaldot Mainnet
- RPC benchmark across public and local nodes
- Batch load planner against observed head data
- Heartbeat worker with Discord / Slack alerts
- Alert-rule test suite and verification scripts
- Public repo, README, and 63-second demo

Next

- Historical charts for block time and extrinsics
- Prometheus export for existing monitoring stacks
- Validator staking and slashing status
- Portaldot testnet deployment as the network opens

PortaPulse is open source under MIT and already deployed: **naomi197.github.io/portapulse** · **github.com/naomi197/portapulse**

THE ASK

Support PortaPulse as the operator toolkit for Portaldot

Node health decides whether a network is usable. PortaPulse gives validators and infrastructure teams the telemetry, benchmarking, and alerting to keep Portaldot nodes reliable — and it is ready to extend as the ecosystem grows.

naomi197.github.io/portapulse · github.com/naomi197/portapulse